# Comparative Analysis of Machine Learning Algorithms and Ensemble Techniques for Diverse Image Classification Tasks


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***Abstract*— In the era of burgeoning image data, the demand for robust image classification algorithms has never been more pressing. This research article delves into the realm of image classification, encompassing a comprehensive analysis of diverse datasets and machine learning algorithms, While simultaneously studying the efficiency of ensemble approaches in improving classification performance. The study employs five distinct datasets, spanning medical images, everyday objects, and natural scenery, ensuring a broad spectrum of classification challenges. These datasets include the Skin Cancer ISIC dataset, CIFAR-10, Flowers, Apparel Image, and Brain Tumor dataset. The image classes within these datasets vary significantly, presenting an ideal testbed for assessing algorithmic versatility. Our investigation scrutinizes five machine learning algorithms: Support Vector Classifier (SVC), Random Forest Classifier (RFC), Gradient Boosting Classifier (GBC), K-Nearest Neighbors (KNN), and Gaussian Naive Bayes (GNB). Each algorithm is evaluated individually, followed by a comprehensive ensemble approach involving a Voting Classifier. The results unveil nuanced performance variations across diverse datasets. Notably, RFC and GBC exhibit remarkable accuracy in brain tumor image classification, while KNN demonstrates strengths in classifying apparel images. Ensemble techniques, embodied by the Voting Classifier, harmonize these algorithms, yielding competitive and balanced performance across the datasets. This article contributes valuable insights into the realm of image classification, shedding light on algorithmic strengths and limitations, the efficacy of ensemble techniques, and their applicability to diverse image datasets. These findings hold significance for fields ranging from medical diagnostics to everyday object recognition, paving the way for more precise and versatile image classification solution**



***Keywords—Image Classification, Machine Learning Algorithm, Ensemble Techniques, Diverse Dataset, Performance Evaluation***


## I. INTRODUCTION (*HEADING 1*)

In the midst of the advancement Artificial Intelligence and Machine Learning, the analysis of diverse image datasets [1] has become a cornerstone for numerous applications, from medical diagnostics [2] , to object recognition[3]. In this context, we present a comprehensive study using five diverse multiclass datasets, namely the Skin Cancer ISIC dataset [4] , CIFAR-10 dataset [5], Brain Tumor dataset [6], Apparel Image dataset [7], and Flower dataset [8]. While these datasets encompass a range of domains, including medicine and general object recognition, they collectively provide a fertile ground for research in image classification and pattern recognition.

This research primarily addresses the difficulty of accurately classifying images into several categories within each dataset. This includes distinguishing between various skin cancer types in medical images, recognizing diverse objects in everyday images, and identifying brain tumor presence. Additionally, the apparel and flower datasets offer unique classification tasks, such as identifying different clothing items and classifying various flower species. These tasks collectively represent the problem statement this research aims to address.

This research aims to develop and evaluate machine learning models for accurately classifying images across diverse datasets, addressing unique challenges in medical imaging and object recognition [9], [10], [11]. The research investigates the effectiveness of individual classification algorithms and explores the benefits of ensemble techniques [12] like voting classifiers in improving classification accuracy.

This research tackles image classification challenges using five datasets with unique characteristics. Preprocessing [13], including data cleaning [14] and augmentation [15], may affect generalization. The study focuses on a selection of classification algorithms[16] and does not consider deep learning due to limited computational resources for experimentation.

This research offers a comprehensive analysis of image classification across diverse datasets, highlighting the impact of ensemble techniques, particularly voting classifiers, on accuracy. It also highlights the strengths [17] and limitations [18] of different algorithms and provides a comparative analysis of classification performance, providing valuable information for selecting the most suitable approaches in image classification scenarios.

This research article is organized to ensure a clear and systematic flow of information. It begins with an introductory section that outlines the background, objectives, research


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questions, and contributions. The section on “Data and Methodology” provides details on the datasets, preprocessing procedures, and classification methods used. The "Experimental Results" section presents and analyzes the findings, followed by the "Discussion" section, which interprets these results relative to the study’s aims and hypotheses. In conclusion, the final section summarizes key findings and suggests possible avenues for future research.

## II. METHODS

This study used a systematic methodology to examine the efficacy of heterogeneous ensemble methods [19] in improving classification accuracy for challenging image datasets. The methodology encompassed data collection, preprocessing, algorithm election, ensemble techniques, and performance evaluation as in Figure 1.

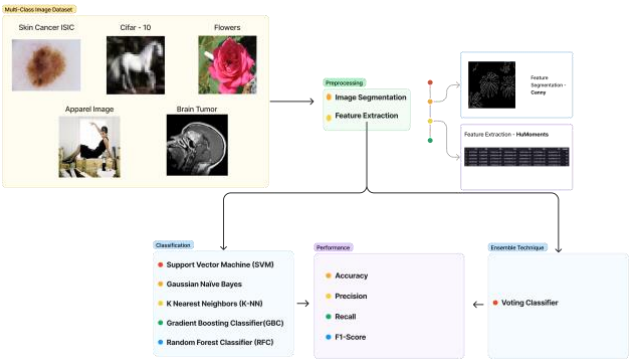


Fig. 1. Research Method

### A. Data Collection

The dataset used was obtained from the Kaggle site. Following the collection of data, the EDA phase is conducted, which entails visualizing the dataset’s attributes, classifications, number of cases, and whether or not there are any missing values. Table 1 displays the dataset’s details. The five datasets under scrutiny are as follows.

*1) Skin Cancer ISIC Dataset :* A repository of dermatological images representing various skin conditions, including melanomas, nevi, squamous cell carcinomas, and more.

*2) CIFAR-10 Dataset :* Comprising a collection of 32x32 color images distributed across ten different classes, CIFAR-10 poses a challenging task for image classification, featuring categories such as automobiles, animals, and natural objects.

*3) Flower Image Dataset :* With sixteen floral categories, ranging from astilbe to water lilies, this dataset challenges the classification model to distinguish between an array of distinct flower types.

*4) Apparel Image Dataset:* Featuring a comprehensive collection of clothing items, this dataset includes categories like dresses, pants, shirts, shoes, and shorts, each available in various colors.

*5)* Brain Tumor MRI Dataset: Focusing on the medical domain, this dataset showcases the potential of classifying brain MRI scans into categories like pituitary tumors, glioma tumors, meningioma tumors, and normal brain scans.

TABLE I. DATASETS INFORMATION

| Dataset | Number of Cases | Number of Attributes | Number of Classes | Attribute Characteristics | Missing Values |
|---|---|---|---|---|---|
| Skin Cancer ISIC | 2475 | 7 | 9 | Numeric | No |
| CIFAR-10 | 60000 | 7 | 10 | Numeric | No |
| Flowers | 15700 | 7 | 16 | Numeric | No |
| Apparel Image | 11384 | 7 | 24 | Numeric | No |
| Brain Tumor | 21700 | 7 | 4 | Numeric | No |

### B. Preprocessing

To prepare the data for modeling, image segmentation using the Canny method was applied for edge detection [20]. Feature extraction was performed using Hu moments, yielding numerical descriptors from the images. The data was then split into attributes (features) and target variables.

*1) Canny Edge Detection*

Canny edge detection [21] is typically based on convolution and gradient operations, and it involves several steps, including Gaussian blurring, gradient computation, non-maximum suppression, and edge tracing. While it might be challenging to represent all the steps mathematically, gradient magnitude as in (1) and gradient direction as in (2).

$$G = \sqrt{(G_x)^2 + (G_y)^2} \tag{1}$$

$$GG_xG_y\theta = \arctan\left(\frac{G_y}{G_x}\right) \tag{2}$$

The vector magnitude is denoted by $G$. The vector component in the x-direction is denoted by$G_x$. The vector component in the y-direction is denoted by $G_y$. $\theta$ represents the angle that the vector makes with the positive x-axis. $G_x$ is the component of the vector in the x-direction. $G_y$ is the component of the vector in the y-direction.

*2) Humoments Feature Extraction*

Humoments [22] are a set of seven moments that are used to describe the shape of an object in an image. These moments are calculated from the central moments of an image. The central moment's is calculated as in (3).

$$\mu_{pq} = \sum_x \sum_y (x - \bar{x})^p (y - \bar{y})^q I(x, y) \tag{3}$$

where $\bar{x}$ and $\bar{y}$ are the centroid coordinates of the image region and $I(x, y)$ is the pixel intensity at coordinates $(x, y)$.

### C. Classification

Five prominent machine learning algorithms were selected to serve as base models for ensemble techniques. The algorithm is Support Vector Machine [23], Random Forest Classifier [24], K Nearest Neighbors [25], Gaussian Naïve Bayes [26], and Gradient Boosting Classifier [27].These algorithms were chosen for their wide applicability and varying strengths in handling diverse datasets.

### 1) *Support Vector Machine*

Support Vector Machines are sophisticated classification algorithms that try to locate the hyperplane that best divides data points from distinct classes [23]. SVM seeks to maximize the margin between different classes while minimizing errors in classification. Utilizing kernel functions, it can address both linear and non-linear classification tasks by mapping the data into higher-dimensional spaces. The primary goal of SVM is to identify the ideal hyperplane that distinctly divides data points into separate categories. The decision function for the radial basis function (RBF) kernel is as follows [28], is as follows (4):

$$f(x) = \text{sign}\left(\sum_{i=1}^{n_{sv}} \alpha_i y_i K(x, x_i) + b\right) \quad (4)$$

Where, $n_{sv}$ is the number of support vectors, $\alpha_i$ are the Lagrange multipliers obtained from the optimization, $y_i$ is the class label $(\pm 1)$, $K(x, x_i)$ is the RBF kernel function : $K(x, x_i) = exp(-\gamma \parallel x - x_i \parallel^2)$ , where $\gamma$ is the kernel coefficient (gamma) , $b$ is the bias term. Figure 2 demonstrates the application of SVM to both binary and multi-class classification problems.

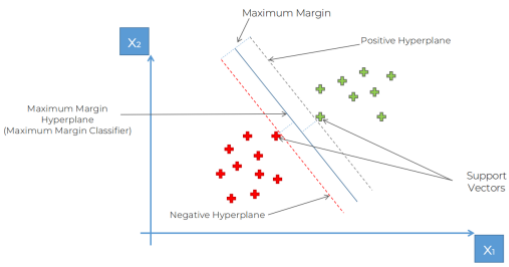


Fig. 2. Support Vector Machine

### 2) *Random Forest Classifier*

As an ensemble method, Random Forest aggregates outputs from several decision trees to improve predictive accuracy[24]. Each tree in the forest is trained on a randomly selected portion of the dataset, with a distinct set of features chosen for training. The final prediction is then obtained by either taking a majority vote or averaging the predictions from each tree, which helps to reduce overfitting and enhances generalization. Consider a dataset with $N$ samples and $M$ features, where each sample is represented as $x_i$ and its associated label as $y_i$. The final prediction is obtained for classification tasks by selecting the most frequently predicted class across all the individual trees. A simplified mathematical outline of the prediction process is as follows: let $K$ denote the total number of trees in the random forest, where $\hat{y}_k(x)$ represents the prediction made by the $k$-th decision tree for the input $x$. For a classification problem with classes $c_i$ , the final prediction of the random forest can be represented as shown in equation (5).

$$P(y = c_i \mid x) = (1 / K) * \Sigma^k \delta(f_k(x), c_i) \quad (5)$$

$P(y = c_i \mid x)$ represents the probability of input x belongs to class $c_i$. $\delta(f_k(x), c_i)$ is the Kronecker delta function, which is 1 if $f_k(x) = c_i$ and 0 otherwise. Random Forest is a versatile method capable of solving a wide range of classification challenges, similar to SVM's usefulness in binary and multi-class classification problems, as illustrated in Figure 3.

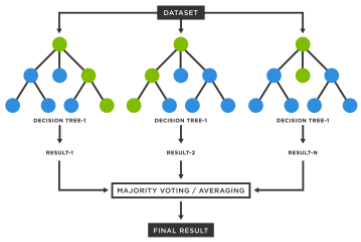


Fig. 3. Random Forest Classifier

### 3) *Gaussian Naïve Bayes*

Gaussian Naïve Bayes is a probabilistic classifier that relies on Bayes' theorem and assumes feature independence [26]. It is ideal for text and image classification tasks. Despite the "naive" assumption of independence, Gaussian Naive Bayes can perform well in practice and is computationally efficient. The predicted class, denoted as $\hat{y}$, is determined using the following formula (6):

$$\hat{y} = \text{argmax}_y \, P(y) \prod_{i=1}^{n} P(x_i | y) \quad (6)$$

Figure 4 depicts the probabilistic decision-making process in Gaussian Naïve Bayes, where the model estimates the class $\hat{y}$ by considering the prior probability of the class, $P(y)$, and the likelihood of the features given the class, $P(x_i \mid y)$.

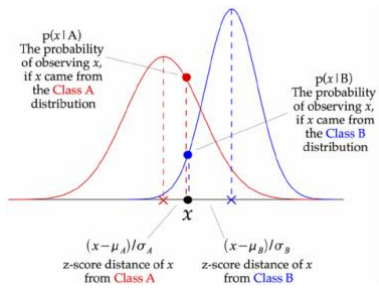


Fig. 4. Gaussian Naïve Bayes

### 4) *Gradient Boosting Classifier*

Gradient Boosting is an ensemble method that strengthens predictive accuracy by aggregating the outputs of multiple weak models, typically decision trees, into a robust model [27]. It works by sequentially training new models, each designed to address errors made by the preceding models. It is well known that Gradient Boosting can make accurate predictions and handle complex relationships in data. Here's a simplified representation of the prediction process in mathematical terms: Let $f^0(x)$ be the initial prediction (e.g., the mean of the target variable), Let $f_k^{-1}(x)$ be the prediction of the model after $(k - 1)$ iterations, Let $h_k(x)$ be the prediction of the $k - th$ weak learner (decision tree). Let $\gamma$ be the learning rate. The updated model prediction after k iterations as shown in (7):

$$f_k(x) = f_k^{-1}(x) + \gamma * h_k(x) \quad (7)$$

In terms of the classification problem, the final prediction can be represented as in (8):

$$P(y = c_i \mid x) = \text{softmax}([f_k(x) \text{ for all classes}]) \quad (8)$$

Here, softmax functions as the activation mechanism, transforming raw model outputs into class probabilities. The iterative Gradient Boosting process, which combines the outputs of several weaker models, generally decision trees, to build a robust predictive model, is illustrated in the schematic presented in Figure 5

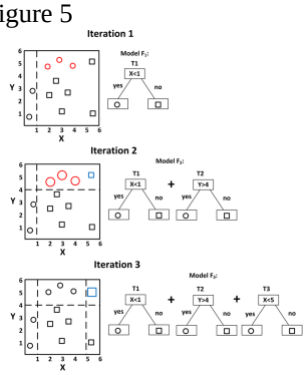


Fig. 5. Gradient Boosting Classifier

*5) K Nearest Neighbour*

A straightforward instance-based learning technique is K Nearest Neighbors(KNN) [25]. KNN classifies a new data point by referencing the labels of its $K$ closest neighbors within the training set. This non-parametric algorithm operates without any assumptions regarding the data's underlying distribution. The distance measure and the number of neighbors (K) can have an impact on how well a KNN performs.

$$\hat{Y} = \text{argmax}_y = \sum_{i=1}^{k} [y_i = y] \quad (9)$$

Equation 9 supplemented by Figure 6, which shows K Neares Neighbors in action. In figure 6 $\hat{y}$ stands for the anticipated class and $y_i$ for the $i$th neighbor's class label.

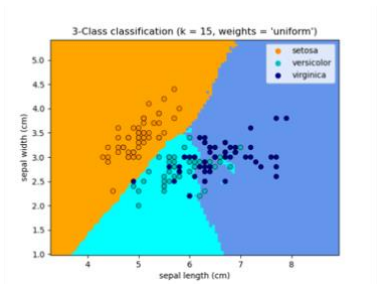


Fig. 6. K Nearest Neighbour

## D. Ensemble Technique

*1) Voting Classifier*

Soft voting is an approach utilized in ensemble learning [19][29] that involves averaging the predicted probabilities from several classifiers to arrive at a final prediction. In a multi-class scenario, the class with the highest average probability from all classifiers is chosen as the final prediction, with each class's probability averaged over the classifiers. Assuming you have $N$ classes and $M$ classifiers, the soft voting formula for multi-class classification can be represented as , where: Let $pij$ be the predicted probability of classifier $j$ for class $i$, where $i = 1,2,\ldots,N$, and $j = 1,2,\ldots,M$. The averaged probability for class $i$ is given by as in(10):

$$Averaged\ Probability_i = \frac{1}{M}\sum_{j=1}^{M} P_{ij} \quad (10)$$

The final predicted class is then determined by selecting the class with the highest averaged probability (11):

$$Final\ Predicted\ Class = (Averaged\ Probability_i) \quad (11)$$

## E. Performance Analysis Comparison

To assess the effectiveness of the ensemble methods, we implemented K-fold cross-validation, which ensured a comprehensive evaluation.. The evaluation metrics included accuracy [30], precision [31], recall [32], and F1-Score [33] Together, these metrics provided a holistic view of the models' performance, reflecting both overall accuracy and class-specific measures.

$$Accuracy = \frac{(TP+TN)}{(TP+TN+FP+FN)} \quad (12)$$

$$Precision = \frac{TP}{(TP+FP)} \quad (13)$$

$$Recall = \frac{TP}{(TP+FN)} \quad (14)$$

$$F1 = \frac{2 \times (Precision \times Recall)}{(Precision + Recall)} \quad (15)$$

In this context, the number of instances accurately identified as positive is known as TP (True Positives), while the instances correctly identified as negative are referred to as TN (True Negatives). Cases that are misclassified as positive when they are, in fact, negative (Type I error) are labeled as FP (False Positives). In contrast, instances that are mistakenly classified as negative despite being positive (Type II error) are referred to as FN (False Negatives).

# III. RESULT AND DISCUSSION

## A. Performance of Individual Classifiers

Tables II, III, IV, V, and VI present the performance results of five distinct classifiers across five different datasets: Support Vector Machine (SVM), Random Forest (RF), Gradient Boosting (GB), k-Nearest Neighbors (k-NN), and Gaussian Naïve Bayes (GNB).

TABLE II. PERFORMANCE RESULT OF SVC

| ∑ *Mean* | *Skin Cancer ISIC* | *Cifar-10* | *Flowers* | *Apparel Image* | *Brain Tumor* |
|---|---|---|---|---|---|
| *Accuracy* | 0.22 | 0.10 | 0.08 | 0.13 | 0.29 |
| *Precision* | 0.14 | 0.05 | 0.04 | 0.09 | 0.19 |
| *Recall* | 0.22 | 0.10 | 0.08 | 0.13 | 0.29 |
| *F1-Score* | 0.14 | 0.02 | 0.02 | 0.07 | 0.13 |

The table indicates that SVC obtains the best accuracy of 0.2949 on the Brain Tumor dataset; however, it typically

exhibits lower values for precision, recall, and F1-score across all datasets, suggesting that there is room for enhancement in the algorithm's ability to accurately classify images in these categories.

TABLE III. PERFORMANCE RESULT OF RFC

| ∑ *Mean* | *Skin Cancer ISIC* | *Cifar-10* | *Flowers* | *Apparel Image* | *Brain Tumor* |
|---|---|---|---|---|---|
| *Accuracy* | 0.31 | 0.17 | 0.16 | 0.19 | 0.38 |
| *Precision* | 0.14 | 0.15 | 0.14 | 0.18 | 0.40 |
| *Recall* | 0.31 | 0.17 | 0.16 | 0.19 | 0.38 |
| *F1* | 0.27 | 0.15 | 0.13 | 0.16 | 0.35 |

The table displays RFC performs exceptionally well in the Brain Tumor dataset, achieving the highest accuracy (0.3808). It also demonstrates balanced precision and recall across datasets, indicating its effectiveness in accurately classifying images into various categories. Overall, RFC exhibits strong potential for diverse image classification tasks, with robust accuracy and precision.

TABLE IV. PERFORMANCE RESULT OF GBC

| ∑ *Mean* | *Skin Cancer ISIC* | *Cifar-10* | *Flowers* | *Apparel Image* | *Brain Tumor* |
|---|---|---|---|---|---|
| *Accuracy* | 0.29 | 0.16 | 0.15 | 0.18 | 0.37 |
| *Precision* | 0.27 | 0.15 | 0.14 | 0.17 | 0.37 |
| *Recall* | 0.29 | 0.16 | 0.14 | 0.18 | 0.37 |
| *F1* | 0.23 | 0.15 | 0.13 | 0.17 | 0.36 |

The Gradient Boosting Classifier (GBC) performance table presents GBC excels in the Brain Tumor dataset, achieving the highest accuracy (0.3750), and it also performs strongly in the Skin Cancer ISIC dataset (0.2901). Moreover, GBC maintains a balanced precision and recall across datasets, demonstrating its effectiveness in accurately classifying images in various categories. Overall, GBC shows promise as a reliable classifier for diverse image classification tasks, particularly excelling in brain tumor image recognition.

TABLE V. PERFORMANCE RESULT OF KNN

| ∑ *Mean* | *Skin Cancer ISIC* | *Cifar-10* | *Flowers* | *Apparel Image* | *Brain Tumor* |
|---|---|---|---|---|---|
| *Accuracy* | 0.26 | 0.13 | 0.10 | 0.14 | 0.35 |
| *Precision* | 0.21 | 0.13 | 0.11 | 0.13 | 0.35 |
| *Recall* | 0.26 | 0.13 | 0.10 | 0.14 | 0.35 |
| *F1* | 0.21 | 0.13 | 0.10 | 0.12 | 0.35 |

While the Brain Tumor dataset achieves the highest accuracy at 0.3596, the performance metrics for the K-Nearest Neighbors (K-NN) classifier suggest that the values for precision, recall, and F1-score could all be enhanced. This indicates that further improvements are needed for better performance in accurately classifying images across the various categories.

TABLE VI. PERFORMANCE RESULT OF GNB

| ∑ *Mean* | *Skin Cancer ISIC* | *Cifar-10* | *Flowers* | *Apparel Image* | *Brain Tumor* |
|---|---|---|---|---|---|
| *Accuracy* | 0.23 | 0.16 | 0.10 | 0.11 | 0.29 |
| *Precision* | 0.17 | 0.16 | 0.10 | 0.17 | 0.25 |
| *Recall* | 0.23 | 0.16 | 0.10 | 0.11 | 0.29 |
| *F1* | 0.17 | 0.11 | 0.06 | 0.09 | 0.19 |

The performance table for the Gaussian Naive Bayes (GNB) classifier shows that, although GNB achieves its highest accuracy of 0.2987 on the Brain Tumor dataset, there is potential for improvement in precision, recall, and F1-score metrics for all datasets. This suggests a need for enhancement to improve its effectiveness in accurately classifying images across these diverse categories.

### *B. Performance Results of Ensemble Learning (Voting Classifier)*

TABLE VII. PERFORMANCE RESULT OF VOTING CLASSIFIER

| ∑ *Mean* | *Skin Cancer ISIC* | *Cifar-10* | *Flowers* | *Apparel Image* | *Brain Tumor* |
|---|---|---|---|---|---|
| *Accuracy* | 0.25 | 0.16 | 0.14 | 0.17 | 0.36 |
| *Precision* | 0.24 | 0.16 | 0.13 | 0.18 | 0.36 |
| *Recall* | 0.24 | 0.16 | 0.14 | 0.17 | 0.35 |
| *F1* | 0.23 | 0.15 | 0.12 | 0.16 | 0.34 |

The performance table (VII) for the Voting Classifier summarizes five different datasets: Skin Cancer ISIC, CIFAR-10, Flowers, Apparel Image, and Brain Tumor. Throughout all datasets, the Voting Classifier demonstrates competitive values achieving its highest accuracy of 0.3600 on the Brain Tumor dataset. This indicates its effectiveness in categorizing images into various classes and highlights its strengths as a robust ensemble method that leverages the benefits of multiple classifiers.

SVC showed moderate performance across various datasets, with the highest accuracy on the Brain Tumor MRI dataset and the lowest on the Cifar-10 dataset. RFC consistently outperformed SVC, with accuracy ranging from 16.31% to 38.08%. GBC displayed competitive performance, achieving accuracy between 15.02% and 37.50%. KNN consistently delivered moderate accuracy, but may not excel in intricate datasets like Cifar-10. GNB displayed mixed performance, achieving moderate accuracy across datasets but struggling with intricate ones. GNB's simplicity made it suitable for clear patterns but limited it in more complex contexts. Overall, SVC demonstrated moderate performance but struggled with diverse object categories.

The ensemble learning approach through the Voting Classifier showcased notable improvements in classification accuracy across all datasets compared to individual algorithms. While individual algorithms struggled to surpass 40% accuracy, the Voting Classifier achieved accuracy levels between 14.37% (Cifar-10) and 36.00% (Apparel Image). The ensemble approach capitalized on the diversity of the algorithms, offering a more robust and balanced classification performance.

Each classification algorithm employed in this study—Support Vector Classifier (SVC), Random Forest (RF), Gradient Boosting (GB), k-Nearest Neighbors (k-NN), and Gaussian Naïve Bayes (GNB)—has its own limitations that affected their performance across various datasets. While the SVC is effective for smaller datasets, it faces challenges with computational efficiency and struggles to manage larger datasets like CIFAR-10 due to its complexity. The Random Forest classifier, although generally robust, can overfit when a large number of trees are used, and its performance may

suffer in the presence of noisy data. Gradient Boosting, a strong ensemble technique, tends to be computationally intensive and often requires careful hyperparameter tuning to avoid overfitting. The primary limitation of k-NN is its sensitivity to irrelevant features, along with slower prediction times for larger datasets. Lastly, GNB operates under the assumption of feature independence, which is seldom accurate in real-world applications, thereby limiting its effectiveness on more complex datasets like those utilized in this study. This research outlines these limitations to offer a well-rounded view of how each algorithm performs in various image classification tasks.

The ensemble approach, implemented through a voting classifier that combined all five algorithms, demonstrated its strengths by balancing individual weaknesses, leading to a more stable and improved accuracy across most datasets. The ensemble method effectively mitigated the biases of each classifier, allowing for better generalization to unseen data. For instance, while individual models like SVC and GNB showed limitations in their accuracy and F1 scores on datasets like CIFAR-10 and the Apparel Image dataset, the ensemble approach achieved more consistent performance by leveraging the strengths of the more accurate classifiers like RFC and GBC. This collaborative decision-making process of the voting classifier enhanced the robustness of predictions and reduced variance. However, the ensemble method also comes with its own limitations, such as increased computational complexity and longer training times due to the need to run multiple models simultaneously. In spite of these difficulties, the ensemble method yielded favorable outcomes, leading to a significant improvement in classification accuracy in contrast to the results achieved by the individual models.

The complexity of datasets significantly impacts classification accuracy. Algorithms like RFC and GBC are effective for complex datasets with diverse object categories. Ensemble learning, like the Voting Classifier, enhances classification accuracy. Choosing an algorithm depends on domain-specific knowledge. Simpler algorithms like GNB are suitable for well-defined datasets, while complex ones like RFC and GBC are better for intricate and diverse ones. Researchers should explore ensemble learning techniques for enhanced classification accuracy.

## CONCLUSION

This study explores multiclass classification across five datasets, focusing on the strengths and limitations of different classification algorithms and the effectiveness of ensemble learning techniques. The complexity and diversity of the dataset affect how well classification systems work. Simpler algorithms, such as Gaussian Naive Bayes, are well-suited for basic tasks, whereas more complex algorithms like Random Forest and Gradient Boosting are effective at managing intricate and varied datasets. Ensemble learning, as demonstrated by the Voting Classifier, significantly improves classification accuracy by leveraging the combined strengths of multiple algorithms. The choice of classification algorithm should be guided by domain-specific knowledge and dataset characteristics, with each algorithm having its strengths and weaknesses. Aligning the choice with the specific application is crucial for optimal classification performance. Classification models can be made much more accurate and robust by using ensemble learning approaches. The analysis emphasizes the dynamic nature of multiclass classification and the importance of considering algorithm diversity, dataset complexity, and domain expertise. As machine learning evolves, the fusion of advanced algorithms and ensemble learning approaches promises even more promising results.

## ACKNOWLEDGMENT

This study is made possible by the Integrated Computer Laboratories (ICLABS) of the department of Computer Science at Universitas Muslim Indonesia

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